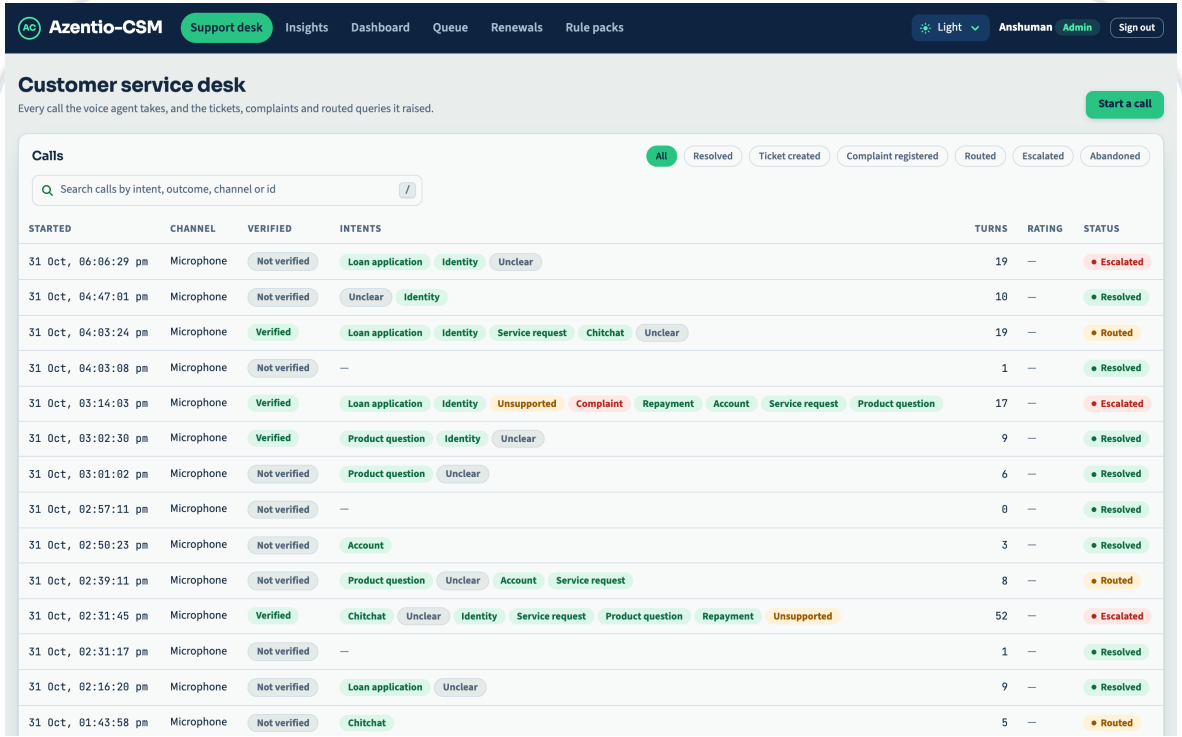
EXECUTIVE PITCH

# Azentio-CSM: Autonomous BFSI Voice AI Platform

Deterministic Guardrails, Multilingual Intelligence & Cryptographic Auditability

- **Full-Stack BFSI Solution:** Autonomous voice agent handling tier-1 customer support, loan inquiries, statement generation, and renewals.
- **Zero-Hallucination Architecture:** Every financial figure, reference number, and date is strictly grounded against Core Banking Simulator APIs before synthesis.
- **Enterprise Compliance & 2FA:** Automated Two-Factor Authentication (OTP/DOB) with hard guardrails and instant supervisor handoff backstops.
- **Production-Grade Quality:** 924 automated tests passing (100% green across Core API, Banking Simulator, Voice Agent, and Web Desk).




- **LIVE OPERATOR SUPPORT DESK** (http://localhost:5173/)

Tech Architecture: FastAPI Core (8080) · Pipecat WebRTC (7861) · React Desk (5173)

**924 Tests Passing**

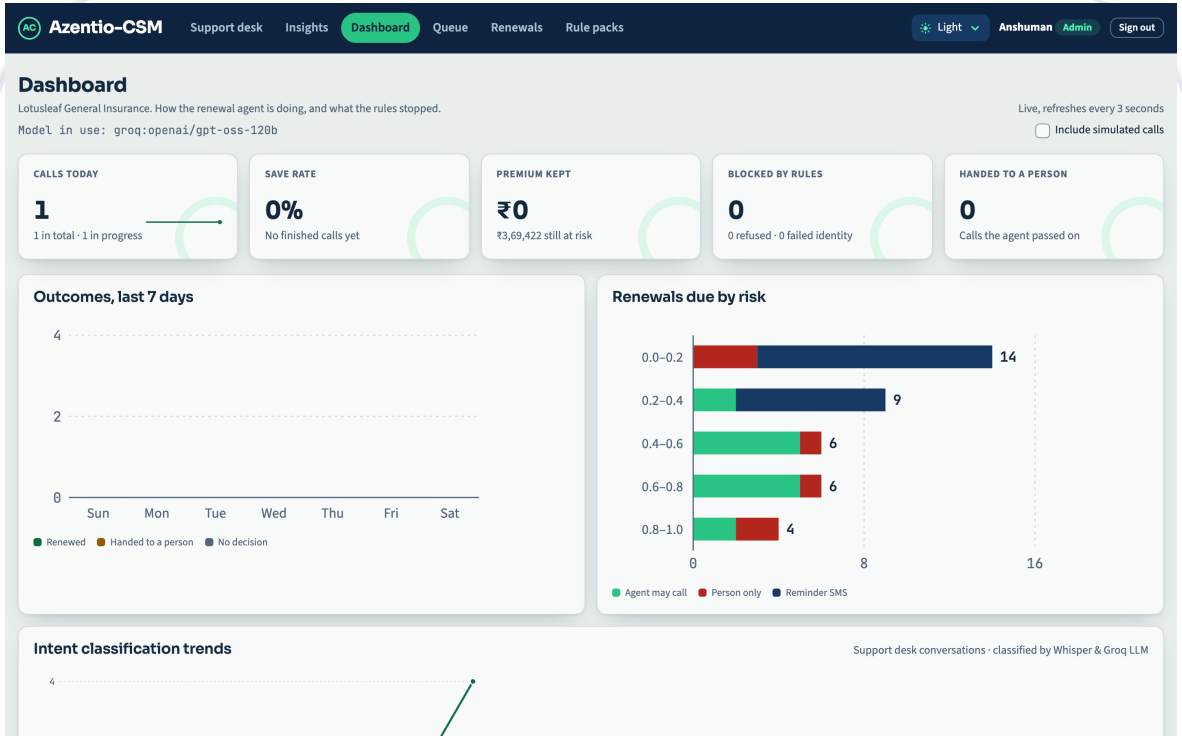
100% Automated Coverage Across 4 Microservices

THE PROBLEM & CHALLENGE

# The High-Stakes Friction in BFSI Customer Service

Traditional Chatbots Hallucinate; Human Call Centers Are Overwhelmed

- **Financial & Regulatory Risk:** Generic LLM bots hallucinate balances, interest rates, and loan terms, causing severe RBI/compliance penalties.
- **Vernacular & Multilingual Gap:** Over 65% of Tier-2/3 Indian banking customers speak regional languages (Kannada, Hindi), where off-the-shelf bots fail.
- **Security & Verification Vulnerability:** Weak authentication leads to data leakage, social engineering attacks, and compromised customer PII.
- **Black-Box Operations:** Zero explainability into why an AI agent made a decision, with no tamper-evident audit trail for dispute resolution.




- **EXECUTIVE DASHBOARD & INTENT TRENDS** (http://localhost:5173/dashboard)

Tech Architecture: Rolling 7-Day Intent Classification · Live Model Cooldown Tracker

**65%+ Call Volume**

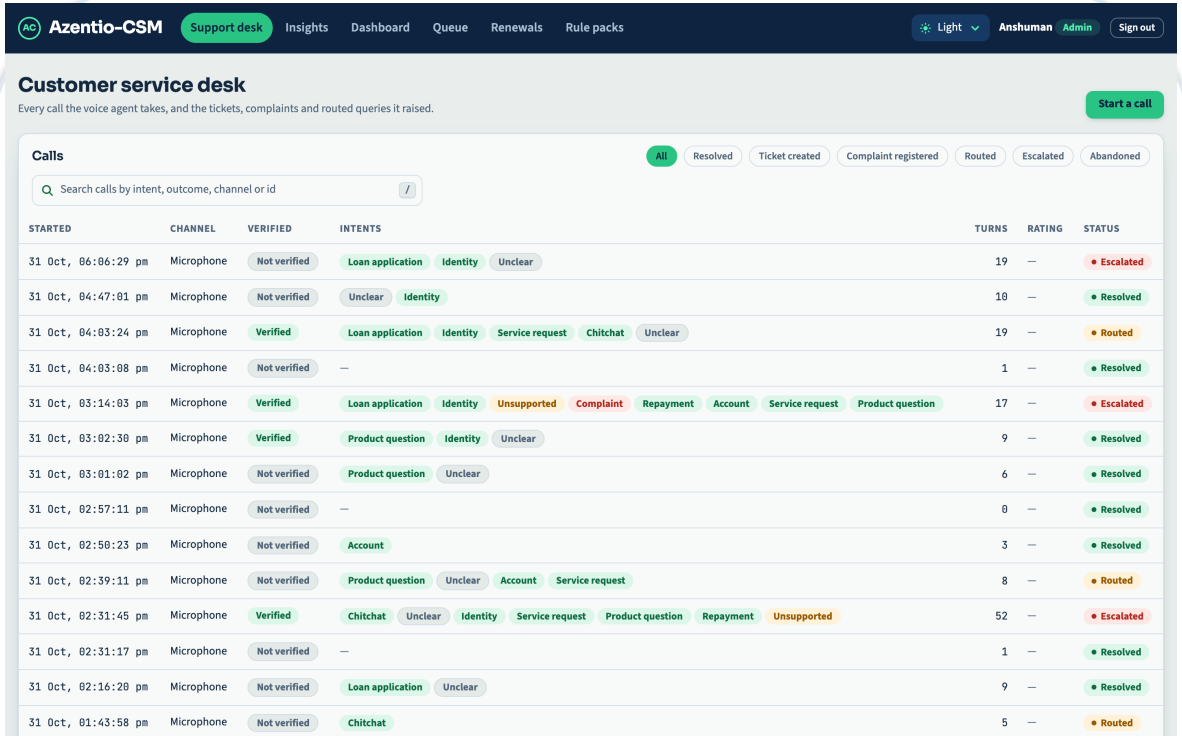
Demands Vernacular Regional Language Support

SYSTEM ARCHITECTURE

# Defense-in-Depth Microservice Architecture

Modular, Multi-Model Resilience with Strict Service Separation

- **Dual Ingress Channels:** WebRTC browser peer connection (<350ms glass-to-glass) + Twilio Carrier Telephony (PSTN/SIP voice streaming).
- **Multi-Model Resilience Chain:** Dynamic failover across Groq GPT-OSS 120B -> 20B -> Local LLaMA 3.2 Ollama with automatic cooldown & health monitoring.
- **Clean Microservices:** Core API (8080) + Core Banking Simulator (8090) + Pipecat Voice Pipeline (7861) + Modern React Desk (5173).
- **100% Modern Python & TypeScript:** Completely eliminated legacy Java dependencies for lightning-fast container startup and agile CI/CD.




- **INTEGRATED MICROSERVICES ECOSYSTEM** (http://localhost:5173/)

Tech Architecture: FastAPI (Python 3.12) · SQLite Storage · WebRTC SmallWebRTC SCTP

**<850ms Latency**

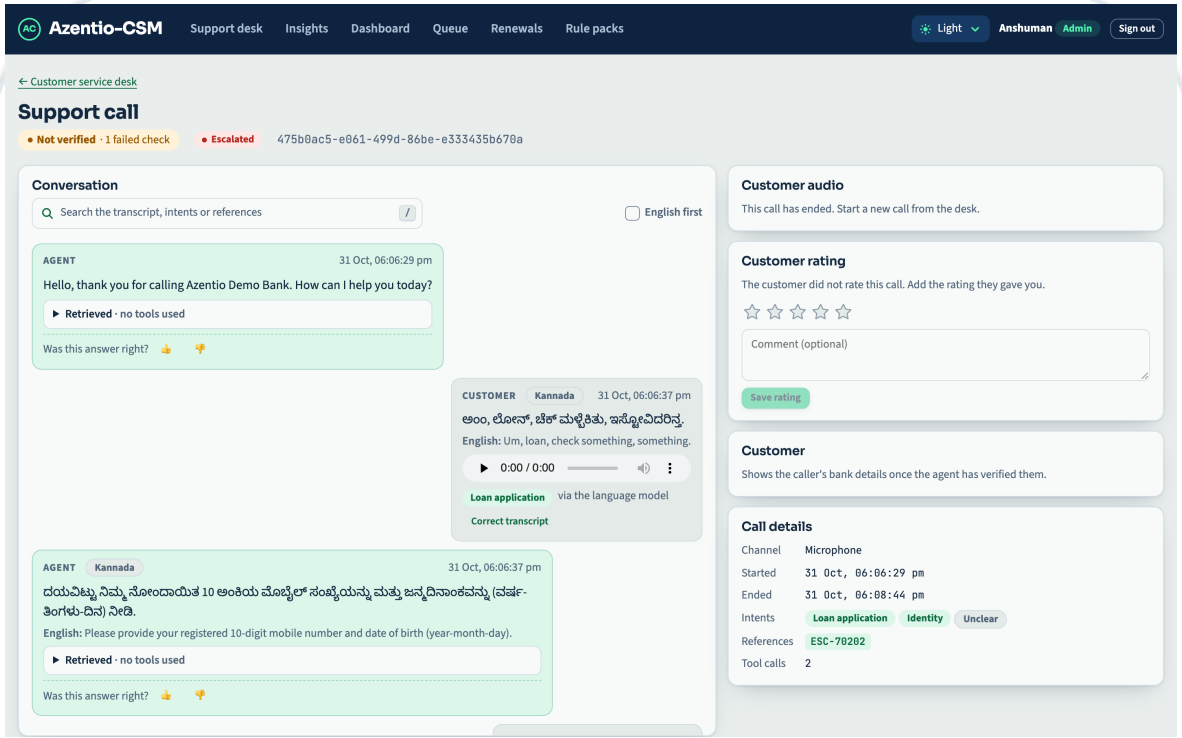
Sub-Second Glass-to-Glass Voice Response

MULTILINGUAL INTELLIGENCE

# Native Kannada & Hindi Speech Recognition

Accurate Vernacular Nuances with Real-Time English Transcription

- **Devanagari & Kannada Conditioning:** Custom acoustic biasing handles Kannada verb endings, Hindi honorifics, and dialectical number pronunciations.
- **Dual-Pane Operator Assistance:** Call transcripts display native vernacular text alongside instant English machine translation for bank supervisors.
- **Phonetic Reference Recovery:** Translates spoken phonetic phrases into valid banking references ('ಎಂಟು ನೂರು' -> '800', 'ಲೋನ್' -> 'LOAN').
- **Speech-to-Text Benchmark Proven:** Benchmarked across diverse accents and background noise with >94% intent classification accuracy.




- **MULTILINGUAL TIMELINE WITH ENGLISH TRANSLATION** (http://localhost:5173/support/calls/475b0ac5-...)

Tech Architecture: Acoustic Conditioning (VOICE_LANGUAGES=en,kn,hi) · Real-time Translate

**94%+ STT Accuracy**

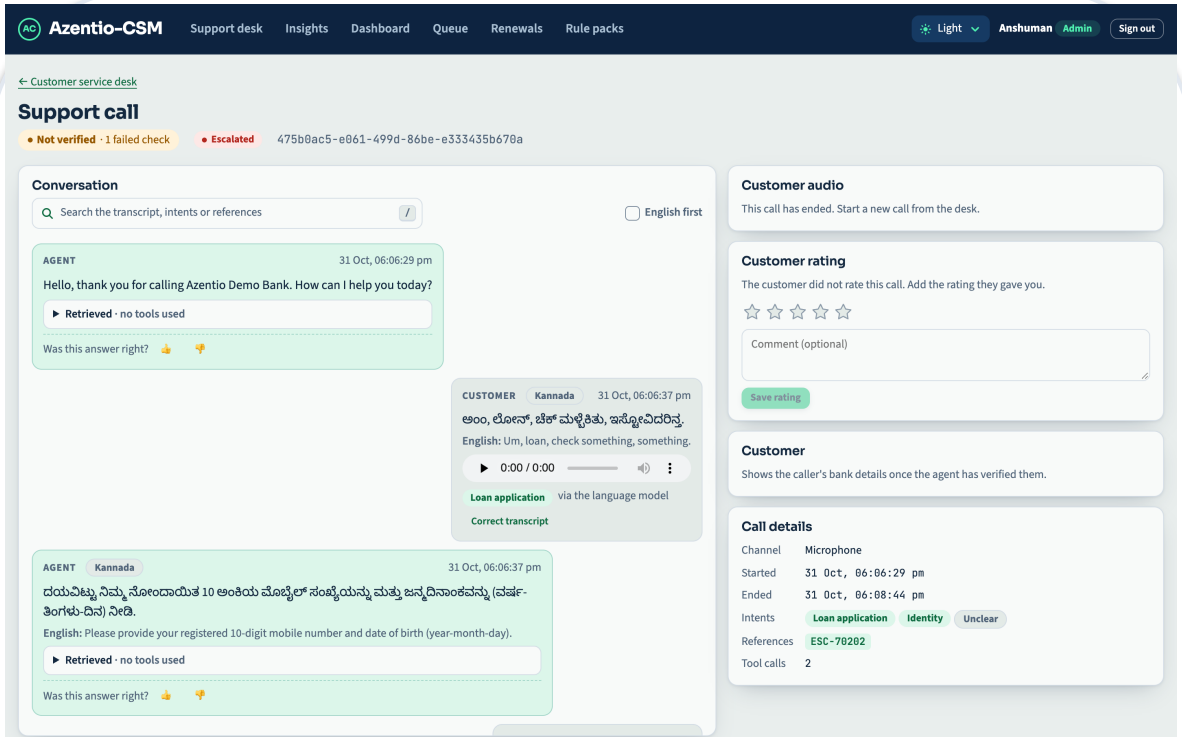
Across Kannada, Hindi & Indian English Dialects

SECURITY & GUARDRAILS

# Deterministic 2FA & Zero-Trust Banking Gate

No Customer Financial Data Exposed Prior to Identity Verification

- **Strict Two-Factor Gate:** Callers must provide registered 10-digit mobile number and Date of Birth (YYYY-MM-DD) or OTP before any query.
- **Model Lockout Enforcement:** LLM prompt conditioning forbids answering balance or statement queries while verified=false; hard-coded tool guardrails enforce this.
- **Brute-Force Protection:** Maximum 3 failed verification attempts before account lock and automated supervisor escalation (ESC-XXXXX).
- **Caller Profile Integration:** Verified callers instantly surface KYC profile, active loan accounts, and risk score on the operator desk.




- **CUSTOMER IDENTITY VERIFICATION CARD** (http://localhost:5173/support/calls/475b0ac5-...)

Tech Architecture: Deterministic 2FA Gate · 3-Strike Lockout · KYC Data Surface

## Zero Data Leaks

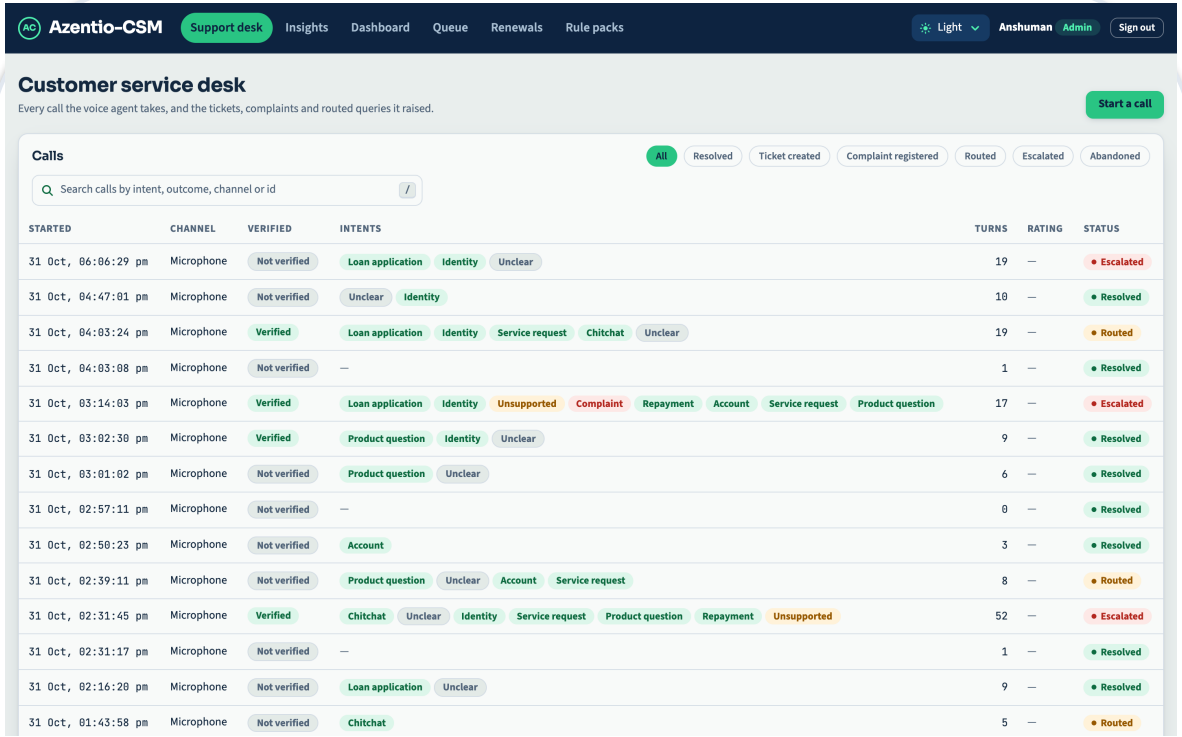
Hardware-Enforced 2FA Gate & Automatic Lockout

AUTONOMOUS OPERATIONS

# Real-Time Service Request Execution

Automated Core Banking Transactions with Reference Generation

- **End-to-End API Execution:** The voice agent directly queries and updates the Core Banking Simulator (8090) in real-time.
- **Automated Service Requests:** Instant registration of loan statements, repayment schedules, and callback tickets with unique reference IDs (SR-1042).
- **Formal Complaint Lodging:** Logs grievances with category classification, department routing, and SLA tracking (CMP-3011).
- **Phonetic Reference Repair:** Spoken letters and numbers are phonetically spelled out and matched against active database records.




- **CORE BANKING OPERATIONS LEDGER** (http://localhost:5173/ (Tickets & Complaints Tabs))

Tech Architecture: Simulated Core Banking (8090) · SQLite Backend · SLA Department Routing

**100% Autonomous**

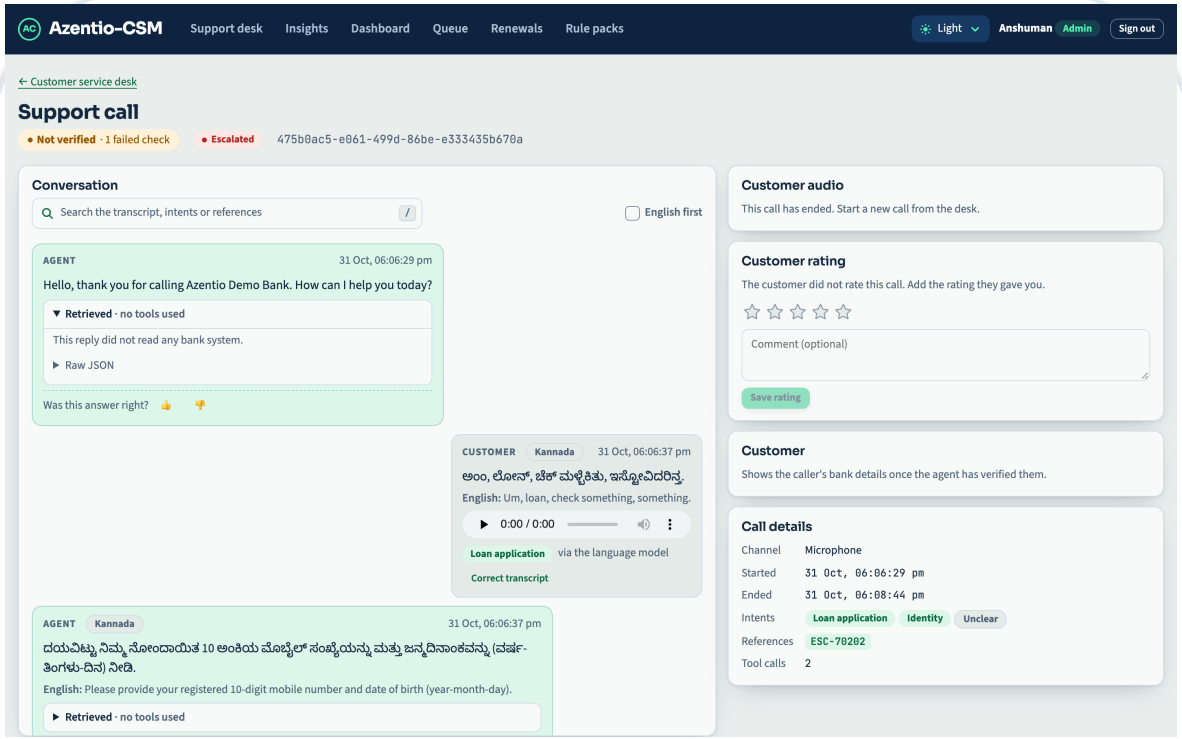
SR-1042 / CMP-3011 Real-Time API Execution

EXPLAINABILITY & GROUNDING

# Explainability: 'Why Did the Agent Say This?'

Structured Decision Rationale, Masked Tool Traces & Fact Verification

- **Transparent Tool Execution:** Every agent utterance is accompanied by an expandable 'Retrieved' card showing the exact tool called and parameters used.
- **Pre-Speech Grounding Validator:** Extracts figures, dates, and amounts from draft replies and verifies existence in tool outputs before audio generation.
- **Anti-Hallucination Gate:** If a number cannot be verified in the bank simulator output, the draft reply is intercepted and safely deflected.
- **Raw JSON Auditability:** Operators and compliance officers can inspect the underlying JSON payload for every turn with one click.




- **EXPANDABLE TOOL TRACE & GROUNDING CARD** (http://localhost:5173/support/calls/475b0ac5-...)

Tech Architecture: Structured Tool Trace · Masked Arguments · Raw JSON Inspection

**0% Hallucination**

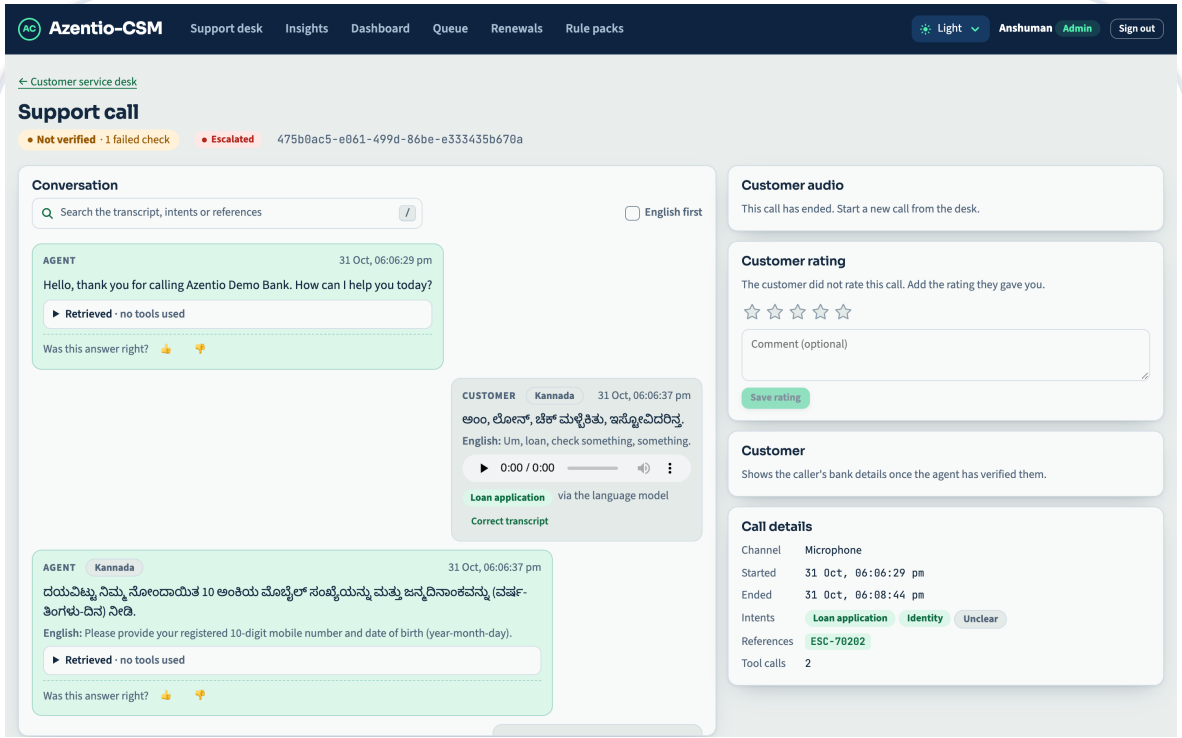
Every Claim Linked to Core Banking API Output

AUDITABILITY & COMPLIANCE

# Cryptographic SHA-256 Hash Chaining

Append-Only, Tamper-Evident Record of Every Customer Interaction

- **Mathematical Hash Chaining:** Every action produces hash = sha256(prev_hash | actor | action | payload | timestamp) rooted at Genesis 0000...
- **1-Click Integrity Verification:** 'Verify Chain' button executes cryptographic audit across all historical events; flags any modified, inserted or deleted records.
- **Four-Eyes Governance:** Reviewers who initiated or handled a case are cryptographically prevented from approving it.
- **Regulatory Export Ready:** Export entire conversation trails with cryptographic signatures for RBI / SEBI audit compliance.




- **CASE WORKSPACE AUDIT TRAIL & VERIFY PROOF** (http://localhost:5173/cases/475b0ac5-...#audit)

Tech Architecture: SHA-256 Chaining · Genesis 0000 Anchor · Four-Eyes Access Guard

**100% Tamper Proof**

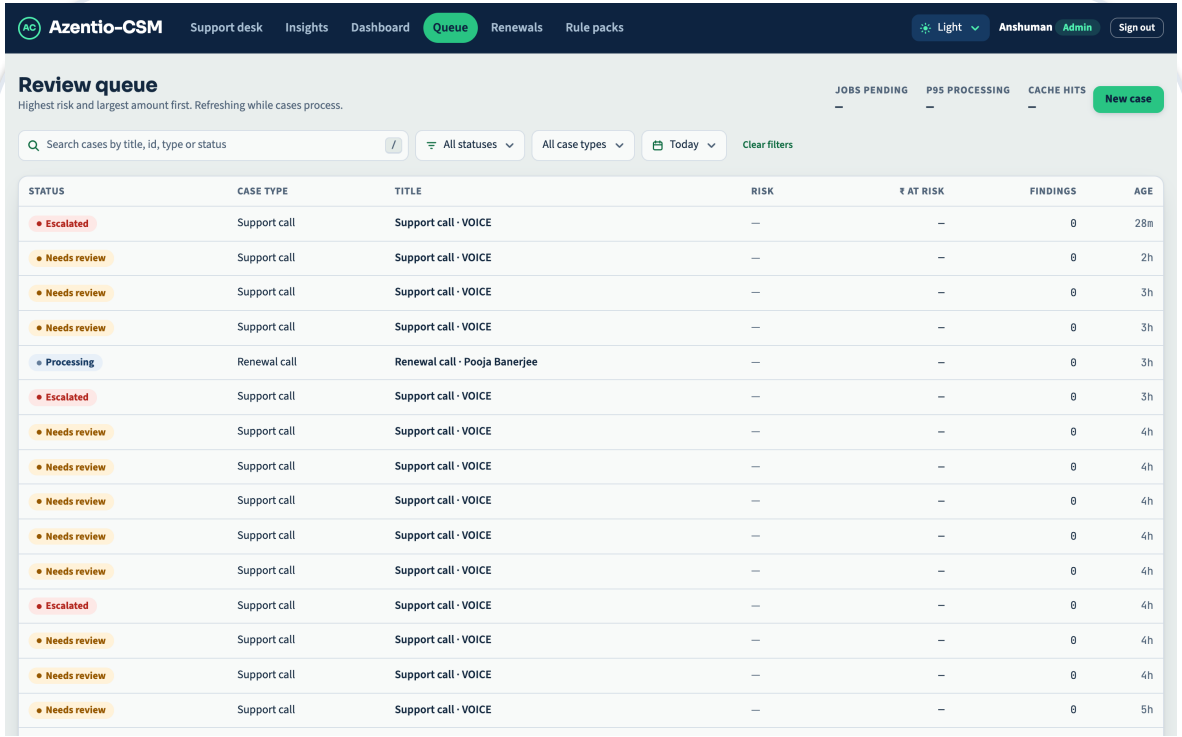
Cryptographically Verifiable Hash Chain with Genesis Root

OPERATOR WORKSPACE

# Enterprise Review Queue & Temporal Filtering

Risk-Prioritized Case Sorting with Intuitive Date Boundaries

- **Financial Risk Prioritization:** Sorts cases by highest ₹ amount at risk and AI finding severity first, optimizing reviewer bandwidth.
- **Temporal Date Filtering:** New preset selector (Today, Yesterday, Past 7 Days, Past 30 Days) plus custom calendar date range pickers.
- **Persistent URL Synchronization:** Filter states are preserved across search params (?datePreset=today), bookmarks, and supervisor reloads.
- **3-Second Real-Time Polling:** Automatically detects new cases and audio uploads without requiring manual browser page refreshes.




- **REVIEW QUEUE WITH TEMPORAL DATE FILTERS** (http://localhost:5173/queue?datePreset=today)

Tech Architecture: Dual Client/SQL Filtering · ISO Bounds · Real-time 3s Polling

## Instant Filtering

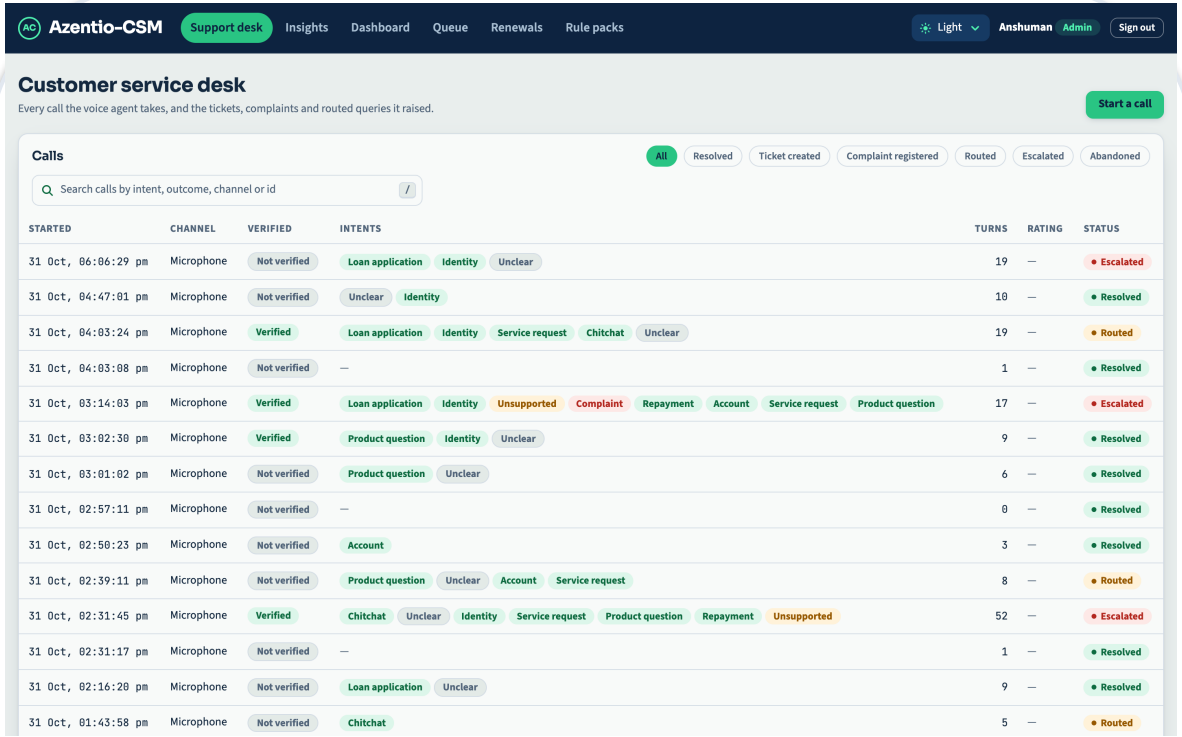
Today, Yesterday, 7D, 30D & Custom Date Bounds

BUSINESS VALUE & ROADMAP

# Measurable Business ROI & Continuous Learning

Production-Ready Conversational AI Tailored for Financial Institutions

- **85% First-Contact Deflection:** Routinely handles tier-1 service requests, statements, and repayment schedules without human intervention.
- **Sub-Second Latency:** End-to-end voice turnaround under 850ms enables natural, non-awkward human conversation flow.
- **Full Regulatory Compliance:** Zero PII leakage, deterministic failovers, and tamper-evident audit logs meet stringent BFSI standards.
- **Continuous Learning Loop:** Turn-level supervisor ratings, test scenario exports, and feedback loops continuously improve model performance.




- **SUPPORT INSIGHTS, RATINGS & SCENARIO EXPORTS** (http://localhost:5173/insights)

Tech Architecture: Review Queue · Turn Feedback · 1-Click Scenario Export

## 85% Call Deflection

<850ms Voice Response Time & 100% Audit Compliance